

Jessie Yuan

Pittsburgh, PA | jzyuan@andrew.cmu.edu | jessie-yuan.github.io

Education

Carnegie Mellon University (CMU) – Pittsburgh, PA

Aug. 2022 – May 2026

B.S. in Computer Science, Minor in Mathematics

- GPA: 4.0/4.0
- Relevant Coursework: Robot Learning (16-831), Fundamentals of Human-Robot Interaction (16-701), Machine Learning (10-701), Embodied AI Safety (16-886), Algorithm Design and Analysis (15-451)

Research Experience

Interactive and Trustworthy Robotics (Intent) Lab – Robotics Institute, CMU

May 2025 – Present

Undergraduate Researcher, Advised by Prof. Andrea Bajcsy

- Implemented a policy steering system that samples candidate action sequences from a base policy, predicts their outcomes with a world model, and uses a VLM verifier to select the action best satisfying the user's instruction.
- Quantified the VLM verifier's uncertainty via conformal prediction (CP) to separate semantic task uncertainty from low-level action feasibility, enabling selection among three resolution strategies: execute when confident, ask a clarifying question, or solicit human corrections to retrain the base policy.
- Designed a CP non-conformity score function inspired by Bayesian models of human intent, decoupling intent inference from behavior evaluation.
- Built a residual learning pipeline for user correction of base policy failures and iterative improvement from collected corrections, outperforming EnsembleDagger and human-gated DAgger baselines.

Robotic Caregiving and Human Interaction (RCHI) Lab – Robotics Institute, CMU

Feb. 2023 – May 2025

Undergraduate Researcher, Advised by Prof. Zackory Erickson

- Developed a ChatGPT-driven speech interface for a commercial feeding robot through prompt engineering.
- Designed a ROS-based system to collect, label, and process data from wearable sensors; cleaned, visualized, and performed feature extraction on this data; and trained and deployed binary classification and regression models using PyTorch.
- Designed, organized, and conducted three human studies, including procedure documentation, participant interaction, and data analysis to collect training data and evaluate system efficacy and user experience.

Publications

J. Yuan*, Y. Wu*, and A. Bajcsy, “When to Act, Ask, or Learn: Uncertainty-Aware Policy Steering”, to appear in *Robotics: Science and Systems (RSS)*, 2026.

A. Padmanabha*, **J. Yuan***, T. Mehta, R. K. Jenamani, E. Hu, V. de León, A. Wertz, J. Gupta, B. Dodson, Y. Yan, C. Majidi, T. Bhattacharjee[†], and Z. Erickson[†], “WAFFLE: A Wearable Approach to Bite Timing Estimation in Robot-Assisted Feeding”, in *The ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2026. **Best Systems Paper.**

A. Padmanabha*, **J. Yuan***, J. Gupta, Z. Karachiwalla, C. Majidi, H. Admoni, and Z. Erickson, “VoicePilot: Harnessing LLMs as Speech Interfaces for Physically Assistive Robots”, in *The ACM Symposium on User Interface Software and Technology (UIST)*, 2024. **Oral Presentation.**

J. Yuan, J. Gupta, A. Padmanabha, Z. Karachiwalla, C. Majidi, H. Admoni, and Z. Erickson, “Towards an LLM-Based Speech Interface for Robot-Assisted Feeding”, in *Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (UIST Adjunct)*, 2024. **Live Demonstration.**

Honors & Awards

Graduate Research Fellowship

2026

National Science Foundation (NSF)

President's Fellowship

2026

Georgia Institute of Technology

Best Systems Paper Award	2026
The ACM/IEEE International Conference on Human-Robot Interaction (HRI)	
Summer Undergraduate Research Fellowship	2023
Carnegie Mellon University	

Teaching Experience

Teaching Assistant for 21-128: Mathematical Concepts and Proofs, <i>CMU</i>	Fall 2024
Teaching Assistant for 15-251: Theoretical Computer Science, <i>CMU</i>	Spring 2024, Fall 2023
Teaching Assistant for 15-122: Imperative Computation, <i>CMU</i>	Spring 2023